BIOCHEMISTRY

DNA/RNA/Protein

1. Molecular biology tools and techniques:
   a. Cloning – introduction of pieces of DNA into a vector in order to permit amplification. Many methods of cloning exist. Commonly, total cellular DNA is cleaved, and each piece is inserted into a vector. The library of vectors is introduced into bacteria or another replication host. A bacteria which has a vector will then replicate, making many copies of the DNA in that vector, hence, a clone. (Lippincott page 404)
   b. cDNA libraries – complementary DNA libraries are made by reverse transcribing (making DNA from RNA) all of the mRNA in a cell. The DNA copies are replicas of mRNA without introns. These can be used as probes, primers, or many other uses.
   c. PCR-see page 146-biochemistry.
   d. Restriction length fragment polymorphism—within the natural sequence of many genes are restriction sites, specific sequences cleaved by restriction enzymes. Many of these sites are polymorphic. That is, they contain differences which render them susceptible or perhaps not susceptible to cleavage at that particular site. These differences can be used to identify genes, or match DNA from different samples, as in forensics.
   e. Sequencing – The major method of sequencing is the Sanger dideoxy nucleotide method. An elongation reaction is carried out using a primer just upstream of the portion to be sequenced. The mixture includes radioactive nucleotides except one of the nucleotides (A,T,G,or C) is dideoxy. That is, it does not have an oxygen at the 2 or 3 position of the ribose sugar. When one of the dideoxy bases is incorporated into the growing chain, elongation is stopped. A reaction is run containing dideoxy nucleotides of each base. For instance, one reaction contains all of the nucleotides but the adenosines are dideoxy. The results are then run on a gel. The sequence can be read by observing which bases were the terminating base at each position on the sequence.
2. Transcriptional regulation
   a. The operon model – This model is related to the regulation of gene transcription (making RNA from genes) in certain environments. For instance the Lac operon consists of regulatory proteins that control production of proteins necessary to degrade lactose. These are only needed when lactose is present.
   b. Eukaryotic transcription—Eukaryotic transcription is controlled by regions of DNA called promoters upstream from the material of the genes. Transcription factors bind to promoters, and help recruit RNA Polymerase II, which binds the TATA box, located approx. 25 bases upstream from the transcription start site. Another sequence, the CAAT box, is located approx. 40 bases upstream from the TATA box. Enhancer regions are other regions of DNA that bind specific protein that aid in transcription of certain gense. These regions can be located upstream, within the gene, within introns, close, or far from the transcription start site. Before a gene is transcribed a large complex of proteins is formed in the promoter region. The need for this large complex of proteins helps in gene regulation.
   c. Role of steroid hormones – Steroid hormones cross membranes and travel directly to the nucleus of target cells. Bound to their respective receptors, steroid hormones act like transcription factors or enhancer binding proteins. They bind to hormone response elements near the genes they regulate and either enhance or inhibit transcription of those genes.
3. Translation (Protein synthesis)
   a. Translation of mRNA occurs in the cytosol on ribosomes. Ribosomes can be free floating or attached to the ER membrane. Three nucleotides on mRNA encode for on amino acid. The start site is AUG on the RNA. This codes for methionine. Each amino acid is attached to a specific tRNA, which recognized the codon for that particular amino acid. Translation happens in three steps: initiation, elongation, and termination.
      i. Initiation: Initiation involves assembling two ribosomal subunits, the mRNA, GTP, the tRNA with the first amino acid, and initiation factors that facilitate the whole process. The ribosome recognizes specific sequences on the mRNA and assembles the machinery. In bacteria, the first amino acid is N-formyl-methionine, while in eukaryotes it is normally methionine. The first tRNA with the appropriate amino acid is in the P-site of the ribosome, and the next tRNA with its appropriate amino acid arrives to the A-site. A peptide bond is formed between the two amino acids. Initiation factors aid in the setting up of the complex.
      ii. Elongation: Elongation factors help the ribosome move down the mRNA with energy derived from GTP hydrolysis. tRNAs are attached to their amino acids using energy from ATP by specific synthetases for each amino acid and tRNA combination. Each time the ribosome moves down the

mRNA, the nascent polypeptide is moved into the P-site, making room in the A-site for a new tRNA/amino acid pair.

      iii. Termination: Termination occurs when the ribosome meets a termination sequence. Release factors cause the new peptide to be released from the ribosomes, and cause the dissociation of the ribosome complex.

   b. Post translational modification of proteins occurs depending on the final destination and function. Modifications include trimming of proteins to active forms. Insulin for example, is synthesized as a zymogen and cleaved to the active molecule. Covalent alterations are also added to some proteins. These include glycosylation with different sugars, phosphorylation, hydroxylation, or association with coenzymes.

4. Acid-base titration curve of amino acid and proteins
   a. Protons will dissociate from weak acids at a certain pH, depending on the strength of the bond of the dissociable hydrogen. This pH is called the pKa of the acid. The Henderson-Hasselbach equation relates the relative amount of acid and base at a given pH to the pKa of an acid.
   b. Amino acids, since they have a carboxyl group, are weak acids. The pKa of most carboxyl groups is around 2, and the pKa for most amino groups of amino acids is around 9. They are referred to as pKa1 and pKa2 respectively. Some amino acids with an acidic or basic side chain have an additional pKa for the side chain hydrogen ion. See page 12 in Lippincott for examples of titration curves.
   c. Proteins have titration curves as well. However, the carboxyl group and the amino group are the main titratable acids, as well as titratable side chains.
   d. Titratable side chains include the acidic amino acids, aspartate and glutamate, basic amino acids arginine, lysine and histidine. The pKa of histidine is 6.0, so at physiologic pH it is not ionized.
5. Role of SH2 domains
   a. The role of SH2 domains is simple: they bind phosphotyrosine. They are normally found in proteins involved in signal transduction. By binding to phosphotyrosine, they allow signals to be passed from one molecule to another. For instance, some receptors, when bound to a ligand, have tyrosine kinase activity. When these tyrosines are phosphorylated, and SH2 domain of another protein can bind to the cytoplasmic phosphotyrosine. The signal can be passed to other proteins and eventually to the nucleus.

Genetic Errors

1. Inherited hyperlipidemias--

Type	Increased lipoprotein class	Increased Lipid
I	Chylomicrons	Triglycerides
IIa	LDL	Cholesterol
IIb	LDL and VLDL	Cholesterol and triglycerides
III	Remnants	Triglycerides and cholesterol
IV	VLDL	Triglycerides
V	VLDL and chylomicrons	Triglycerides and cholesterol

2. Glycogen and lysosomal storage diseases are covered in First Aid.
3. Porphyrias --
   a. Porphyrias are defects of porphyrin metabolism, leading to buildup of toxic metabolites. Porphyrins are ring structures. An example in heme without the iron. Many defects exist in many different steps in the pathways. They are classified depending on clinical and biochemical properities. Five main types exist:
      i. congenital erythropoietic porphyria
      ii. erythrohepatic porphyria
      iii. acute intermittent porphyria
      iv. porphyria cutanea tarda
      v. mixed porphyria
   b. Manifestations include light sensitivity, vesicles that heal when scarring, anemia. Pathogenesis is not well understood.
   c. The clinical picture is someone with light sensitivity, so they only go out at night. They have weird fluorescing molecules in their bodies, so their teeth fluoresce (porphyrin rings.) Because of defects in heme synthesis and metabolism, anemia can be a problem, so they want to drink blood. Some say that vampire legends came from people with porphyrias. Interesting eh?

4. **DNA repair defects** (First Aid p. 149)

Disease	Features	Type of repair defect
Xeroderma pigmentosum (skin sensitivity to UV light)	Skin tumors, photosensitivity, cataracts, neurological abnormalities	Nucleotide excision repair defects, including mutations in helicase and endonuclease genes
Cockayne syndrome	Reduced stature, skeletal abnormalities, optic atrophy, deafness, photosensitivity, mental retardation	Defective repair of UV-induced damage in transcriptionally active DNA; considerable etiological and symptomatic overlap with xeroderma pigmentosum and trichothiodystrophy
Fanconi anemia	Anemia; leukemia susceptibility; limb kidney, and heart malformations; chromosome instability	As many as eight different genes may be involved, but their exact role in DNA repair is not yet known
Bloom's syndrome (radiation)	Growth deficiency, immunodeficiency, chromosome instability, increased cancer incidence	Mutations in the reqQ helicase family
Werner syndrome	Cataracts, osteoporosis, atherosclerosis, loss of skin elasticity, short stature, diabetes, increased cancer incidence; sometimes described as "premature aging"	Mutations in the reqQ helicase family
Ataxia-telangiectasia (x-rays)	Cerebellar ataxia, telangiectases*, immune deficiency, increased cancer incidence, chromosome instability	Normal gene product is likely to involved in halting the cell cycle after DNA damage occurs
Hereditary nonpolyposis colorectal cancer	Proximal bowel tumors, increased susceptibility to several other types of cancers	Mutation in any of four DNA mismatch repair genes
*Telangectases are vascular lesions caused by the dilatation of small blood vessels. This typically produces discoloration of the skin. Reference: Medical Genetics p. 39		

5. Triplet repeat diseases.

Disease	Description	Repeat sequence	Normal & Abnormal range	Parent in which expansion usually occurs	Location of expansion
Huntington disease	Loss of motor control, dementia, affective disorder	CAG	6 to 34; 36 to >100	More often through father	Exon
Spinal and bulbar musculaar atrophy	Adult-onset motor-neuron disease associated with androgen insensitivity	CAG	11 to 34; 40 to 62	More often through father	Exon
Spinocerebellar ataxia (type 1, 2, 3, 6)	Progressive ataxia and other type specific symptoms	CAG	Varies with type	More often through father	Exon
Dentatorubral-pallidoluysian atrophy/Haw River syndrome	Cerebellar atrophy, ataxia, myoclonic epilepsy, choreoathetosis, dementia	CAG	7 to 25; 49 to 88	More often through father	Exon
Myotonic dystrophy	Muscle loss, cardiac arrhythmia, cataracts, frontal balding	CTG	5 to 37; 100 to >1000	Either parent, but expansion to congenital form through mother	3' Untrans-lated region
Friedreich's ataxia	Progressive limb ataxia, dysarthria, hypertrophic cardiomyopathy, pyramidal weakness in legs	GAA	7 to 22; 200 to 900 or more	Disorder is autosomal recessive, to disease alleles are inherited from both parents	Intron
Fragile X syndrome (FRAXA)	Mental retardation, large ears and jaws, macro-orchidism in males	CGG	6 to 52; 200 to >2,000	Exclusively through mother	5' Untras-lated region
Fragile site FRAXE	Mild mental retardation	GCC	6 to 35; 200 or more	More often through mother	Unknown
Reference: Medical Genetics p. 83					

6. Inherited defects in amino acid metabolism.

Name	Prevalence	Mutant gene product	Chromosomal location
Phenylketonuria (PKU)	1/10,000	Phenylalanine hydroxylase	12q24
Tyrosinemia (type 1)	1/100,000	Fumarylacetoacetate hydrolase	15q23-25

Maple syrup urine disease	1/180,000	Branched-chain ∀-ketoacid decarboxylase (multiple subunits)	Multiple loci
Alkaptonuria	1/250,000	Homogentisic acid oxidase	3q2
Homocystinuria	1/340,000	Cystathionine ∃-synthase	21q2
Oculocutaneous albinism	1/35,000	Tyrosinase	11q
Cystinosis	1/100,000	Unknown	17p
Cystinuria	1/7,000	SLC3A1 (type 1)	2p
Reference: Medical Genetic p.138			

Metabolism

1. **Glycogen synthesis: regulation, inherited defects.**
   - A. **Regulation:** In the well fed state, *Glycogen synthetase* is allosterically activated by glucose 6-phosphate, as well as by ATP, a high energy signal in the cell. An elevated insulin level results in overall increased glycogen synthesis.
     Glucagon (in the liver) and Epinephrine (muscle and liver) bind cell membrane receptors and stimulate *adenylate cyclase* then *cAMP*. *Glycogen synthetase* is then phosphorylated by *cAMP-dependent protein kinase* which inhibits the production of glycogen. (Lippincott's Biochem p. 142-145)
   - B. **Inherited defects: Glycogen storage diseases- First Aid p. 150**
2. **Oxygen consumption, carbon dioxide production, and ATP production for fats, proteins, and carbohydrates.**
   - A. **Oxygen consumption** occurs in the mitochondrial matrix. Cytochrome oxidase uses oxygen as the final electron acceptor and converts it to H_2O.
   - B. **Carbon dioxide production** results from reactions in several pathways including the TCA cycle and the Hexose Monophosphate Pathway (HMP). One molecule of CO_2 (and one NADH) is produced in the conversion of Pyruvate to Acetyl-CoA by *pyruvate dehydrogenase*. Two molecules of CO_2 are produced in the TCA cycle for every molecule of Acetyl-CoA. One molecule of CO_2 (and one NADPH) is also produced by the conversion of 6-Phosphogluconate to Ribulose 5-phosphate in the HMP.
   - C. **ATP production**
     Fats are broken down through Triacylglycerol degradation into fatty acids which are then broken down to Fatty Acyl CoA and then Acetyl-CoA which then enters the TCA cycle. Each Acetyl-CoA produces 3NADH, $1FADH_2$, $2CO_2$, 1GTP which is equal to 12ATP/acetyl CoA.
     Proteins are broken down to amino acids which enter the TCA cycle at different points. Refer to p.244 Fig. 22.22 in Lippincott's Biochem for the metabolism of specific amino acids and associated genetic deficiencies.
     Carbohydrates are broken down to monosaccharides, the most common of which is D-glucose. Aerobic metabolism of glucose produces 38 ATP via malate shuttle, 36 ATP via G3P shuttle. Anaerobic glycolysis produces only 2 ATP per glucose molecule.
3. **Amino acid degradation pathways (urea cycle, tricarboxylic acid cycle).** Refer to Lippincott's Biochem Figure 21.11, p.237 and Figure 22.2, p.244.
4. **Effect of enzyme phosphorylation on metabolic pathways.**

Enzyme	Enzyme Activity when Phosphorylated	Description
Glycogen phosphorylase	Active	Degrades glycogen
Glycogen synthase	Inactive	Synthesizes glycogen
Pyruvate kinase	Inactive	Converts Phosphoenolpyruvate (PEP) to Pyruvate
Pyruvate dehydrogenase	Inactive	Converts Pyruvate to Acetyl-CoA
Acetyl CoA carboxylase	Inactive	Converts Acetyl CoA to Malonyl CoA in triglyceride synthesis
Hormone sensitive lipase	Active	Breaks triglyceride into a fatty acid and a diglyceride

Note: not a complete table- add enzymes as needed		

5. Rate limiting enzymes in different metabolic pathways: First Aid p. 155

6. Sites of different metabolic pathways (What organ? Where in the cell?).
   - -**Organ sites** (First Aid 99 p 156):
     - -**Liver**: Most represented, including gluconeogenesis; fatty acid oxidation (β-oxidation); ketogenesis; lipoprotein formation; urea, uric acid & bile acid formation; cholesterol synthesis.
     - -**Brain**: Glycolysis, amino acid formation.
     - -**Heart**: Aerobic pathways (e.g., β-oxidation and Krebs cycle)
     - -**Adipose tissue**: Esterification of fatty acids and lipolysis
     - -**Muscle**: *fast twitch*: Glycolysis; *slow twitch*: Aerobic pathways
   - -**Cell sites** (First Aid 99 p 154):
     - -**Mitochondria**: β-oxidation, acetyl-CoA production, Krebs cycle.
     - -**Cytoplasm**: Glycolysis, fatty acid synthesis, HMP shunt, protein synthesis (RER), steroid synthesis (SER).
     - -**Both**: Gluconeogenesis, urea cycle, and heme synthesis
7. Fed state versus fasting state: forms of energy used, direction of pathways.
   - -See **liver diagrams** for both states on page 159 of First Aid 99.
   - -**Fed (Absorptive) state** (BRS Biochemistry p 4):
     - -Glucose is oxidized by various tissues for energy or is stored as glycogen in liver and muscle. In liver, glucose is also converted to triacylglycerols, which are packaged in VLDL and released into the blood. Fatty acids of the VLDL and chylomicrons are stored in adipose tissue. Absorbed amino acids are used by various tissues to synthesis proteins, produce nitrogen-containing compounds, and produce energy.
   - -**Fasting state** (BRS Biochemistry p 7):
     - -With decreasing blood glucose level, the liver is stimulated by glucagon to supply glucose (glycogenolysis & gluconeogenesis) and ketones to the blood. The liver uses amino acids from muscles and fatty acids and glycerol from adipose tissue.
   - -**Prolonged Fasting** (BRS Biochemistry p 9):
     - -**Muscles**: ↓ use of ketones & ↑ oxidation of fatty acids for primary energy source.
     - -**Brain**: ↑use of abundant ketones instead of glucose.
     - -**Liver**: ↓ gluconeogenesis & spares muscle proteins.

8. Tyrosine kinases and their effects on metabolic pathways (insulin receptor, growth factor receptors)
   - -**Insulin Receptor** (Lippincott p 273):
     - -Insulin binding activates receptor tyrosine kinase activity in the intracellular domain of the β-subunit.
     - -Tyrosine residues of the β-subunit are autophosphorylated.
     - -Receptor tyrosine kinase phosphorylates other proteins, such as the insulin receptor substrate (IRS).
     - -Phosphorylated IRS promotes activation of other protein kinases and phosphatases, leading to the biological actions of insulin (see Topic 13 below).
   - -**Insulin-like Growth Factor Receptor** (BRS Physiology p 249):
     - -The IGF receptor has tyrosine kinase activity like the insulin receptor

9. Anti-insulin hormones (e.g., glucagon, GH, cortisol).
   - -Considered “counterregulatory hormones” because they oppose many actions of insulin (Lippincott p 275):
     - -**Glucagon**: *Acute, short-term regulation* by stimulating hepatic glycogenolysis and **gluconeogenesis**.
     - -**Epinephrine**: *Acute, short-term regulation* by promoting glycogenolysis and lipolysis, inhibiting insulin secretion, and inhibits the insulin-mediated uptake of glucose by peripheral tissues.
     - -**Cortisol**: *Long-term management* by stimulating **gluconeogenesis** and lipolysis.
     - -**Growth Hormone**: *Long-term management* by stimulating **gluconeogenesis** and lipolysis.

10. Synthesis and metabolism of neurotransmitters.

-**Acetylcholine** (Correlative Neuroanatomy p 30):
  -ACh is synthesized from acetyl-CoA and choline by the enzyme choline acetyltransferase in the presynaptic cholinergic nerve terminal.
  -ACh is broken down after release into the synaptic cleft by the enzyme acetylcholinesterase.
  -Choline is taken back up into the presynaptic nerve terminal to be converted back into ACh.
-**Catecholamines** (Correlative Neuroanatomy p 30):
  -Phenylalanine→tyrosine→DOPA→**dopamine**→**norepinephrine**→**epinephrine** (First Aid 99 p151).
  -Tyrosine is converted to DOPA by tyrosine hydroxlyase.
  -DOPA is converted to dopamine by DOPA decarboxylase.
  -Dopamine is hydroxlyated to NE and NE is converted to epinephrine by phenylethanolamine-*N*-methyltransferase.
  -Dopamine and NE are inactivated by both MAO (presynaptic nerve terminal) and COMT (postsynaptic).
-**Serotonin** (Correlative Neuroanatomy p 32): Synthesized from the amino acid tryptophan.

11. Purine/pyrimidine degradation:
-**Purine** (G, A) degradation (BRS Biochemistry p 265):
  -First **phosphate** and **ribose** are removed; then the nitrogenous base is oxidized.
  -**Guanine** is degraded to **xanthine** and **adenine** to **hypoxanthine**, which is then oxidized to xanthine by xanthine oxidase (this enzyme requires molybdenum).
  -**Xanthine** is oxidized to **uric acid** by xanthine oxidase.
  -The **kidneys** excrete **uric acid**, which is not very water-soluble.
-**Pyrimidine** (C, U, T) degradation (BRS Biochemistry p 267):
  -Unlike the purine rings, which are not cleaved in human cells, the pyrimidine ring can be opened and degraded to a highly soluble structures, such as β-alanine and β-aminoisobutyrate.
  -The carbons produce CO_2 and the nitrogens produce urea.

12. Carnitine shuttle: function and inherited defects.
-**Function** (Lippincott p 182):
  -The carnitine shuttle transports the acyl group from **cytosolic** fatty acyl CoA molecules across the inner mitochondrial membrane, which is *impermeable* to CoA, returning it to **mitochondrial** CoA molecules.
  -The newly formed mitochondrial fatty acyl CoA molecules can then undergo β-oxidation.
-**Inherited defects**:
  -The congenital absence of a carnitine acyltransferase in skeletal muscle, or low concentrations of carnitine due to defective synthesis, result in an inability to use **long-chain fatty acids** as a metabolic fuel, causing myoglobinemia and weakness following exercise.

13. Cellular/organ effects of insulin secretion (Lippincott p 273 and BRS Biochemistry p 154).
-**Liver**: ↑ glycogen synthesis; ↓ glucose production by inhibiting gluconeogenesis & glycogenolysis; ↑ triacylglycerol synthesis & conversion to VLDL.
-**Muscle**: ↑ glycogen synthesis; ↑ glucose uptake by increasing the number of glucose transporters.
-**Adipose tissue**: ↓ triacylglycerol degradation & ↑ triacylglycerol synthesis; ↑ glucose uptake by increasing the number of glucose transporters.
-**Most tissues**: ↑ entry of amino acids into cells & ↑ protein synthesis.
-Insulin does **NOT** significantly stimulate the transport of glucose into tissues such as liver, brain, & RBCs.

14. Effect of uncouplers on oxidative phosphorylation (Lippincott p 71).
-Compounds **that increase the permeability of the inner mitochondrial membrane to protons** can uncouple electron transport and phosphorylation.
-The energy produced by the transport of electrons, without a proton gradient, is released as **heat** rather than being used to synthesize ATP.
-**2,4-dinitrophenol**, a lipophilic proton carrier that readily diffuses through the membrane, is an uncoupler.
-**Aspirin** in high doses (as well other **salicylates**) is an uncoupler. This explains the fever that accompanies the toxic overdoses of these drugs.
-Uncoupling is different that just inhibiting electron transport, like cyanide does.